COLIMITS ALONG A FUNCTOR WITH CONNECTED FIBRES AND LIFTING OF MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2059-2081, following the statement at 2041-2057.

Editorial note. The alternative cofinal-route sentence at proof lines 2060-2062 is retained verbatim; the counted proof is the direct argument that follows it, which is complete with no omission markers.

Allowed standard prerequisites (editorial).

- Fibre category of a functor. definition-fibre-category, cited at statement lines 2046-2047. Supplies the fibre categories of I over I' whose connectedness is assumed in (1) and used at proof lines 2073-2077.
- Elements of a limit as compatible families. the limits of Hom sets displayed at proof lines 2064-2068. Lets both sides of the claimed equality be handled as sets of compatible families of morphisms into the test object T.

Source reference. Original definition 02XH: sources/stacks-categories/pinned-original-source.tex, source lines 6261–6261. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04E7: sources/stacks-categories/pinned-original-source.tex, source lines 1977–1977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{I} \to \mathcal{I}'$ be a functor. Assume*

(1) *the fibre categories (see Definition 02XH) of $\mathcal{I}$ over $\mathcal{I}'$ are all connected, and*
(2) *for every morphism $\alpha' : x' \to y'$ in $\mathcal{I}'$ there exists a morphism $\alpha : x \to y$ in $\mathcal{I}$ such that $F(\alpha) = \alpha'$.*

Then for every diagram $M : \mathcal{I}' \to \mathcal{C}$ the colimit $\mathrm{colim}_{\mathcal{I}} M \circ F$ *exists if and only if* $\mathrm{colim}_{\mathcal{I}'} M$ *exists and if so these colimits agree.*

Proof. One can prove this by showing that $\mathcal{I}$ is cofinal in $\mathcal{I}'$ and applying Lemma 04E7. But we can also prove it directly as follows. It suffices to show that for any object T of $\mathcal{C}$ we have

$$\lim_{\mathcal{I}^{opp}} \mathrm{Mor}_{\mathcal{C}}(M_{F(i)}, T) = \lim_{(\mathcal{I}')^{opp}} \mathrm{Mor}_{\mathcal{C}}(M_{i'}, T)$$

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If $(g_{i'})_{i' \in \mathrm{Ob}(\mathcal{I}')}$ is an element of the right hand side, then setting $f_i = g_{F(i)}$ we obtain an element $(f_i)_{i \in \mathrm{Ob}(\mathcal{I})}$ of the left hand side. Conversely, let $(f_i)_{i \in \mathrm{Ob}(\mathcal{I})}$ be an element of the left hand side. Note that on each (connected) fibre category $\mathcal{I}_{i'}$ the functor $M \circ F$ is constant with value $M_{i'}$. Hence the morphisms f_i for $i \in \mathrm{Ob}(\mathcal{I})$ with $F(i) = i'$ are all the same and determine a well defined morphism $g_{i'} : M_{i'} \to T$. By assumption (2) the collection $(g_{i'})_{i' \in \mathrm{Ob}(\mathcal{I}')}$ defines an element of the right hand side. $\square$